

Credit Card Fraud Detection Using Machine Learning

Student Information

Name: Zunaira Jawaid Pilani

Project Title

Credit Card Fraud Detection Using Machine Learning

Introduction

Credit card fraud is one of the biggest challenges in online financial transactions. Fraud detection systems help banks and financial institutions identify suspicious transactions and reduce financial losses. In this project, machine learning techniques are used to classify transactions as either genuine or fraudulent.

Objectives

- Detect fraudulent credit card transactions.
- Handle imbalanced data using SMOTE.
- Compare Logistic Regression and Random Forest models.
- Evaluate model performance using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC Score.

Dataset

- Dataset: Credit Card Fraud Detection Dataset
- Total Transactions: **284,807**
- Features: **30**
- Target Column: **Class**
 - **0 = Normal Transaction**
 - **1 = Fraudulent Transaction**

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Imbalanced-learn (SMOTE)

- VS Code

Methodology

1. Load the dataset.
2. Perform data exploration.
3. Check missing values.
4. Analyze class distribution.
5. Split data into training and testing sets.
6. Apply feature scaling using StandardScaler.
7. Balance the training data using SMOTE.
8. Train Logistic Regression.
9. Train Random Forest.
10. Evaluate both models using different performance metrics.

Machine Learning Models

Logistic Regression

A simple and efficient classification algorithm used as a baseline model.

Random Forest

An ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Performance Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- ROC-AUC Score

Results

Both models successfully detected fraudulent transactions. Random Forest provided stronger overall performance, while Logistic Regression achieved high fraud recall after balancing the dataset with SMOTE.

Conclusion

This project demonstrates that machine learning can effectively identify fraudulent credit card transactions. Applying SMOTE improved the handling of imbalanced data, and Random Forest

produced highly reliable predictions. Such systems can assist financial institutions in reducing fraud and improving transaction security.

Future Improvements

- Hyperparameter tuning
- XGBoost and LightGBM implementation
- Deep Learning models
- Real-time fraud detection
- Web deployment using Flask or Streamlit